

Problem 2.

(a)

the truth table shows that the $(P \wedge \neg Q) \vee (P \wedge Q)$ is equivalent to P .

P	Q	$\neg Q$	$P \wedge \neg Q$	$P \wedge Q$	$(P \wedge \neg Q) \vee (P \wedge Q)$
T	T	F	F	T	T
T	F	T	T	F	T
F	T	F	F	F	F
F	F	T	F	F	F

(b)

$(A - B)$ is equivalent to $A \wedge \neg B$

the truth table about $(A - B) \vee (A \wedge B)$ is

A	B	$\neg B$	$A \wedge \neg B$	$A \wedge B$	$(A - B) \vee (A \wedge B)$
T	T	F	F	T	T
T	F	T	T	F	T
F	T	F	F	F	F
F	F	T	F	F	F

we can see the A is equivalent to $(A - B) \vee (A \wedge B)$

so $x \in A \iff x \in ((A - B) \vee (A \wedge B))$

Problem 3.

(a) $\text{sixish}(n) := \exists m \exists k. ((6 \cdot 10^k + m = n) \wedge m < 10^k)$

(b) $\exists m \exists n \forall k. Q(m) \wedge Q(n) \wedge k \neq m \wedge k \neq n \wedge \neg Q(k)$

(c) $\forall k \exists n \exists m. (\text{isEven}(k) \wedge k > 2) \implies (m + n = k) \wedge \text{isPrime}(n) \wedge \text{isPrime}(m)$

(d) $\forall n. n > 100 \implies \text{sixish}(n)$

(e) it is not equivalent because when $n = 1$, $R(n)$ should be True but we can't find any $b \in \mathbb{N}$ to make $1 = 2b + 5$

Problem 4.

(a)

We use induction.

Proof. Let $P(n)$ be the statement that if $Q(x_1 \cdot x_2 \cdots x_n)$ is true, then at least one of $Q(x_1), Q(x_2), \dots, Q(x_n)$ must be true.

Base case: For $n = 1$, the statement holds trivially. For $n = 2$, if $Q(x_1 \cdot x_2)$ is true, then at least one of $Q(x_1)$ or $Q(x_2)$ must be true which holds by the given premise.

Inductive hypothesis: Assume that $P(n)$ is true for some $n \geq 1$. That is, if $Q(x_1 \cdot x_2 \cdots x_n)$ is true, then at least one of $Q(x_1), \dots, Q(x_n)$ must be true.

Inductive step: We assume that for $n \geq 1$, $P(n)$ is true. We have to prove $P(n+1)$ is true, which means we have to prove if $Q(x_1 \cdot x_2 \cdots x_{n+1})$ is true, then at least one of $Q(x_1) \dots Q(x_{n+1})$ must be true. Since the $x_1 \cdot \dots \cdot x_{n+1}$ can be written as $(x_1 \cdot \dots \cdot x_n) \cdot x_{n+1}$, so let a be $x_1 \cdot x_2 \cdots x_n$ and b be x_{n+1} , if $Q(a \cdot b)$ is true then one of $Q(a)$ or $Q(b)$ must be true.

- Case 1: if $Q(a)$ is true, then $Q(x_1 \cdots x_n)$ is true by the inductive hypothesis
- Case 2: if $Q(b)$ is true, then $Q(x_{n+1})$ is true which proved the $P(n+1)$ is true directly.

So we proved $P(n)$ is true for all $n \geq 1$.

□

(b)

For part (a), we should choose $Q(x)$ to be the predicate: "x is irrational"

To apply the result from part(a), we have to prove the case of $n = 2$ holds.

Prove it by contrapositive

Proof. $Q(x_1 \cdot x_2) \implies Q(x_1) \vee Q(x_2)$ is equivalent to its contrapositive $\neg Q(x_1) \wedge \neg Q(x_2) \implies \neg Q(x_1 \cdot x_2)$, So we can prove $(x_1 \text{ and } x_2 \text{ are rational})$ IMPLIES $(x_1 \cdot x_2 \text{ is rational})$ directly.

Since x_1 and x_2 are rational, we can let x_1 be $\frac{a}{b}$ and x_2 be $\frac{c}{d}$ ($a, b, c, d \in \mathbb{N}^+$). And $x_1 \cdot x_2$ will be $\frac{a \cdot c}{b \cdot d}$, because both $a \cdot c$ and $b \cdot d$ are integer, the $\frac{a \cdot c}{b \cdot d}$ must be rational, so we proved $\neg Q(x_1 \cdot x_2)$, as required. $\square$

(c)

proof by contradiction.

Proof. Assume the sake of contradiction that $Q(x_1 \cdot x_2) \wedge (\neg Q(x_1) \wedge \neg Q(x_2))$.

That tells us x_1 and x_2 is rational, so we can let $x_1 = \frac{a}{b}$ and $x_2 = \frac{c}{d}$ ($a, b, c, d \in \mathbb{Z}$ with $b, d \neq 0$)

Then $x_1 \cdot x_2$ become to $\frac{a \cdot c}{b \cdot d}$, which is rational. $\Rightarrow \Leftarrow$ $\square$